

Routing benchmark: Solomon instances

T3 · Appendix A1–A2 · 56 instances · 100 customers per instance

Internal benchmark reported by Omelet. Results below are transcribed from the exact final-pitch appendix, not newly rerun tests. They support discussion of experimental performance; they do not establish shipped NVIDIA SDK adoption. Original run logs and named test-owner confirmation are not attached; independent verification is pending.

Metric	Reported gap, instance win count and runtime.
Baseline / comparison	Omelet RouteOpt, cuOpt, RouteOpt-ALNS, LKH-3 and OR-Tools.
Setup	Public Solomon CVRPTW set; 56 × 100-customer instances; nominal 10-second budget. The appendix describes same-instance, same-machine comparisons but does not separately identify the Solomon machine or run configuration.
Gap reference	The full-results appendix defines gap against the best compared solver per instance. Do not label this as a verified gap to an external best-known-solution database.
Test date / owner	Not separately specified for Solomon. Date and responsible engineer/team: not yet verified.

METHOD	HARDWARE	REPORTED GAP	WINS / 56	REPORTED TIME
Omelet RouteOpt	CPU	+0.08%	38	10.0 s
NVIDIA cuOpt	CPU + GPU	+1.76%	13	9.7 s
RouteOpt, ALNS configuration	CPU	+2.53%	3	8.1 s
LKH-3	CPU	+3.41%	2	47.6 s
OR-Tools	CPU	+4.56%	0	10.0 s

Time/feasibility exception: LKH-3 is reported feasible on 52/56 instances and at 47.6 seconds. This row is not an equal 10-second-budget result. Handling of infeasible cases in the aggregate gap is not specified.

Source and reproducibility

[Open appendix A2: full results](#)

Per-instance results, seed/repetition counts, solver settings, feasibility handling and gap aggregation should accompany a reproducible benchmark. Current status: company-reported experiment; not independently verified.